

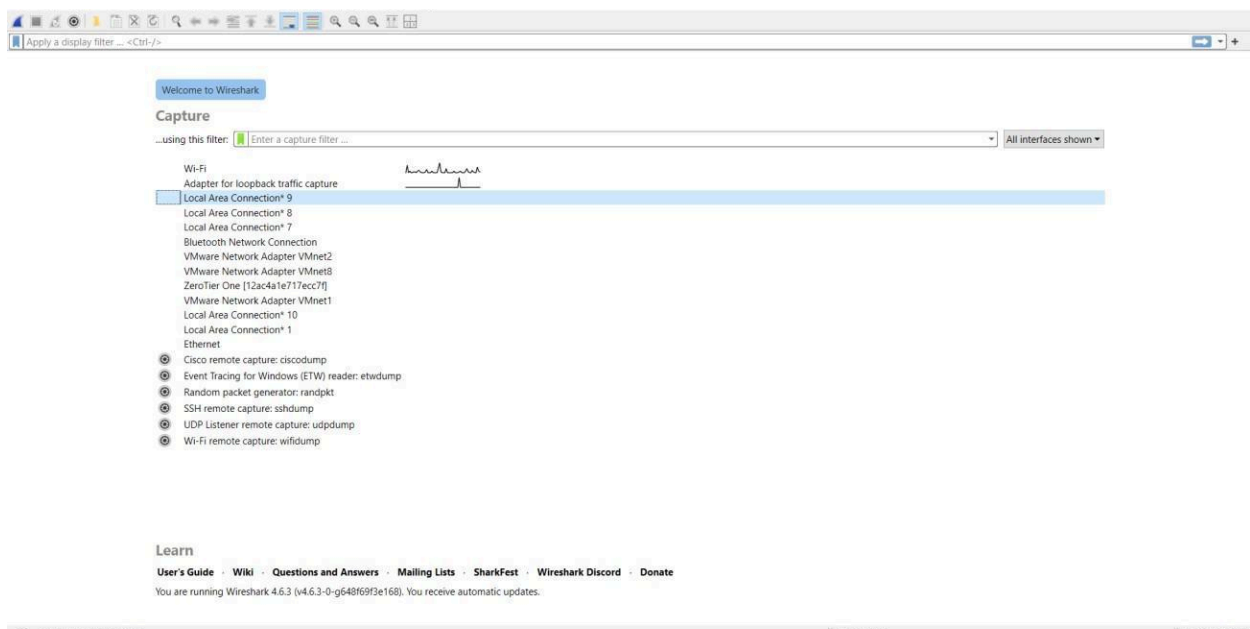
Advanced Computer Networks – PCS223

Assignment 1

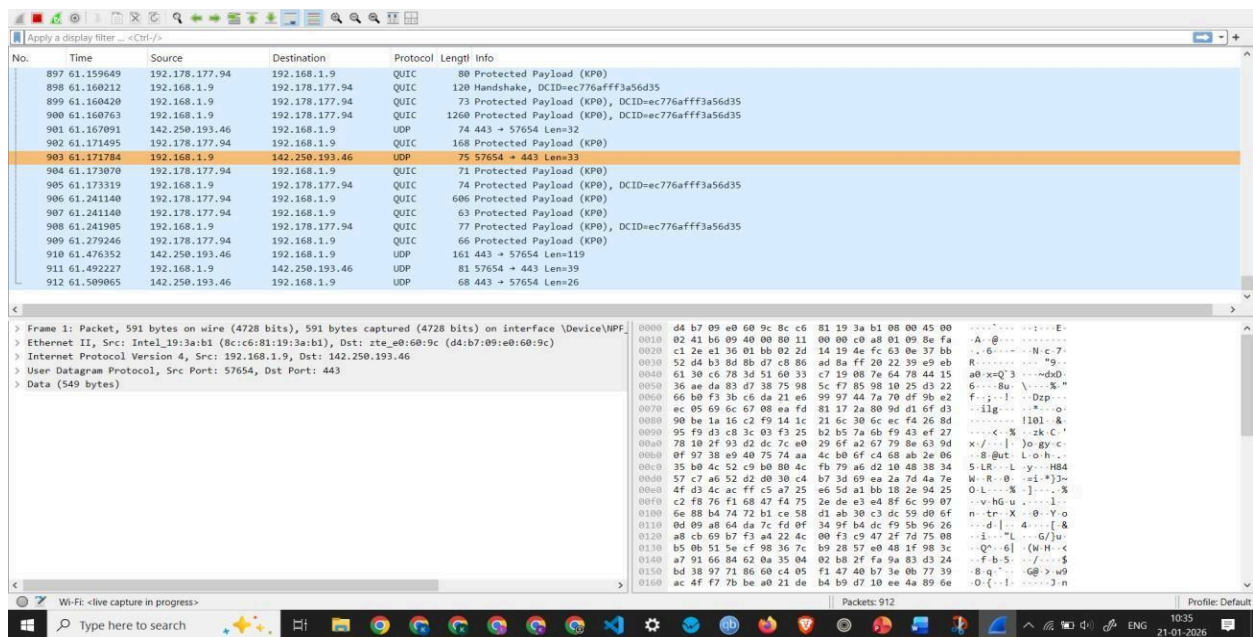
Name – Taranpreet Singh

Subgroup – CS4

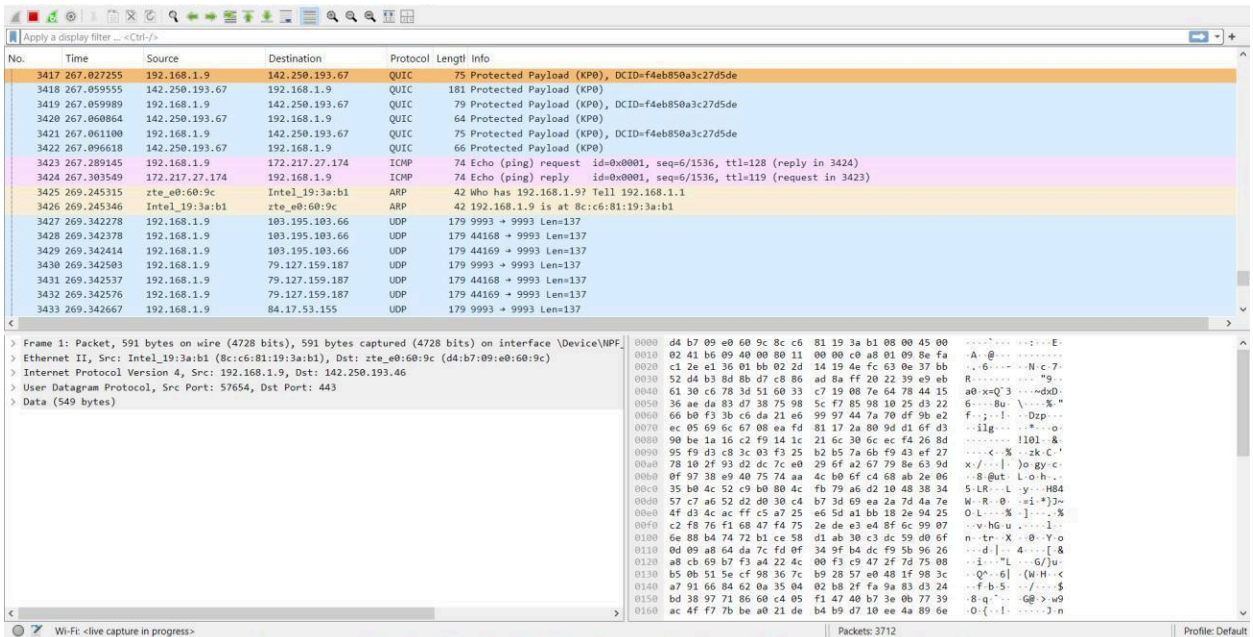
Roll No. – 8025320103



Screenshot showing Wireshark Home screen showing all interfaces and the active interface (Wi-Fi)



Showing Live Packet Capture on Wi-Fi



Screenshot showing ICMP Packets Captured by pinging google.com

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.9	142.250.193.46	UDP	591	57654 → 443 Len=549
2	0.018956	142.250.193.46	192.168.1.9	UDP	73	443 → 57654 Len=31
3	0.029040	192.168.1.9	142.250.193.46	UDP	75	57654 → 443 Len=33
4	0.031408	192.168.1.9	172.64.148.235	TLSv1.2	109	Application Data
5	0.044116	172.64.148.235	192.168.1.9	TCP	56	443 → 64482 [ACK] Seq=1 Ack=56 Win=19 Len=0
6	0.062220	192.168.1.9	142.251.43.100	QUIC	1292	Initial, DCID=ce45a285749b77e3, PKN: 1, PADDING, CRYPTO, PING, CRYPTO, CRYPTO, CRYPTO, PING, CRYPTO, PADDING, PING
7	0.062305	192.168.1.9	142.251.43.100	QUIC	1292	Initial, DCID=ce45a285749b77e3, PKN: 2, CRYPTO
8	0.062337	192.168.1.9	142.251.43.100	QUIC	1292	Initial, DCID=ce45a285749b77e3, PKN: 3, CRYPTO, CRYPTO, CRYPTO, CRYPTO, PADDING, CRYPTO, PADDING, PING, PING, CRYPTO, CRYPTO, CRYPTO
9	0.062594	192.168.1.9	142.251.43.100	QUIC	122	0-RTT, DCID=ce45a285749b77e3
10	0.078984	142.251.43.100	192.168.1.9	QUIC	82	Initial, SCID=ee45a285749b77e3, PKN: 1, ACK
11	0.079868	142.251.43.100	192.168.1.9	QUIC	82	Initial, SCID=ee45a285749b77e3, PKN: 2, ACK
12	0.080460	142.251.43.100	192.168.1.9	QUIC	1292	Initial, SCID=ee45a285749b77e3, PKN: 3, ACK, PADDING
13	0.080460	142.251.43.100	192.168.1.9	QUIC	1292	Initial, SCID=ee45a285749b77e3, PKN: 4, ACK, PADDING
14	0.082092	142.251.43.100	192.168.1.9	QUIC	1292	Initial, SCID=ee45a285749b77e3, PKN: 5, CRYPTO, PADDING
15	0.082092	142.251.43.100	192.168.1.9	QUIC	344	Protected Payload (KP0)
16	0.082092	142.251.43.100	192.168.1.9	QUIC	990	Protected Payload (KP0)
17	0.082092	142.251.43.100	192.168.1.9	QUIC	117	Protected Payload (KP0)

Frame 4: Packet, 591 bytes on wire (4728 bits), 591 bytes captured (4728 bits) on interface \Device\NPF...
 Ethernet II, Src: Intel_19:3a:b1 (8c:c6:81:19:3a:b1), Dst: zte_e0:60:9c (d4:b7:09:e0:60:9c)
 Internet Protocol Version 4, Src: 192.168.1.9, Dst: 142.250.193.46
 User Datagram Protocol, Src Port: 57654, Dst Port: 443
 Data (549 bytes)

Screenshot showing IP Packets

No.	Time	Source	Destination	Protocol	Length	Info
4	0.031408	192.168.1.9	172.64.148.235	TLSv1.2	109	Application Data
5	0.044116	172.64.148.235	192.168.1.9	TCP	56	443 → 64482 [ACK] Seq=1 Ack=56 Win=19 Len=0
153	5.548948	172.64.148.235	192.168.1.9	TLSv1.2	78	Application Data
154	5.549299	192.168.1.9	172.64.148.235	TLSv1.2	82	Application Data
155	5.562645	172.64.148.235	192.168.1.9	TCP	56	443 → 64482 [ACK] Seq=25 Ack=84 Win=19 Len=0
234	20.241400	192.168.1.9	44.213.187.187	TCP	55	64668 → 443 [ACK] Seq=1 Ack=1 Win=510 Len=1
235	20.287794	192.168.1.9	52.203.201.52	TCP	55	64675 → 443 [ACK] Seq=1 Ack=1 Win=513 Len=1
238	20.560284	192.168.1.9	98.94.226.245	TCP	55	64677 → 443 [ACK] Seq=1 Ack=1 Win=512 Len=1
239	20.607690	192.168.1.9	98.94.226.245	TCP	55	64671 → 443 [ACK] Seq=1 Ack=1 Win=512 Len=1
240	20.619240	44.213.187.187	192.168.1.9	TCP	66	443 → 64668 [ACK] Seq=1 Ack=2 Win=267 Len=0 SLE=1 SRE=2
241	20.671944	52.203.201.52	192.168.1.9	TCP	66	443 → 64675 [ACK] Seq=1 Ack=2 Win=128 Len=0 SLE=1 SRE=2
242	20.834755	98.94.226.245	192.168.1.9	TCP	66	443 → 64677 [ACK] Seq=1 Ack=2 Win=128 Len=0 SLE=1 SRE=2
243	20.879899	98.94.226.245	192.168.1.9	TCP	66	443 → 64671 [ACK] Seq=1 Ack=2 Win=327 Len=0 SLE=1 SRE=2
290	25.549188	172.64.148.235	192.168.1.9	TLSv1.2	78	Application Data
291	25.549416	192.168.1.9	172.64.148.235	TLSv1.2	82	Application Data
292	25.560337	172.64.148.235	192.168.1.9	TCP	56	443 → 64482 [ACK] Seq=49 Ack=112 Win=19 Len=0
293	25.979204	192.168.1.9	57.144.41.32	TLSv1.2	124	Application Data

Frame 4: Packet, 109 bytes on wire (872 bits), 109 bytes captured (872 bits) on interface \Device\NPF...
 Ethernet II, Src: Intel_19:3a:b1 (8c:c6:81:19:3a:b1), Dst: zte_e0:60:9c (d4:b7:09:e0:60:9c)
 Internet Protocol Version 4, Src: 192.168.1.9, Dst: 172.64.148.235
 Transmission Control Protocol, Src Port: 64482, Dst Port: 443, Seq: 1, Ack: 1, Len: 55
 Transport Layer Security

Screenshot showing TCP Packets Capture

No.	Time	Source	Destination	Protocol	Length	Info
141	3.377117	192.168.1.9	84.17.53.155	UDP	179	44169 → 9993 Len=137
142	3.377217	192.168.1.9	185.152.67.145	UDP	179	9993 → 9993 Len=137
143	3.377251	192.168.1.9	185.152.67.145	UDP	179	44168 → 9993 Len=137
144	3.377287	192.168.1.9	185.152.67.145	UDP	179	44169 → 9993 Len=137
145	3.522301	79.127.159.187	192.168.1.9	UDP	671	9993 → 9993 Len=629
146	3.552301	84.17.53.155	192.168.1.9	UDP	671	9993 → 9993 Len=629
147	3.645792	185.152.67.145	192.168.1.9	UDP	671	9993 → 44169 Len=629
148	3.652686	183.195.103.66	192.168.1.9	UDP	671	9993 → 44169 Len=629
149	4.971491	192.168.1.9	142.251.43.100	UDP	71	62496 → 443 Len=29
150	5.013433	142.251.43.100	192.168.1.9	UDP	70	443 → 62496 Len=28
156	6.032330	192.168.1.9	172.217.26.106	UDP	71	56825 → 443 Len=29
157	6.076858	172.217.26.106	192.168.1.9	UDP	67	443 → 56825 Len=25
158	6.462845	142.251.43.100	192.168.1.9	UDP	117	443 → 62496 Len=75
159	9.374878	192.168.1.9	183.195.103.66	UDP	70	44169 → 9993 Len=28
160	9.375124	192.168.1.9	79.127.159.187	UDP	179	9993 → 9993 Len=137
161	9.375159	192.168.1.9	79.127.159.187	UDP	179	44168 → 9993 Len=137
162	9.375188	192.168.1.9	79.127.159.187	UDP	179	44169 → 9993 Len=137

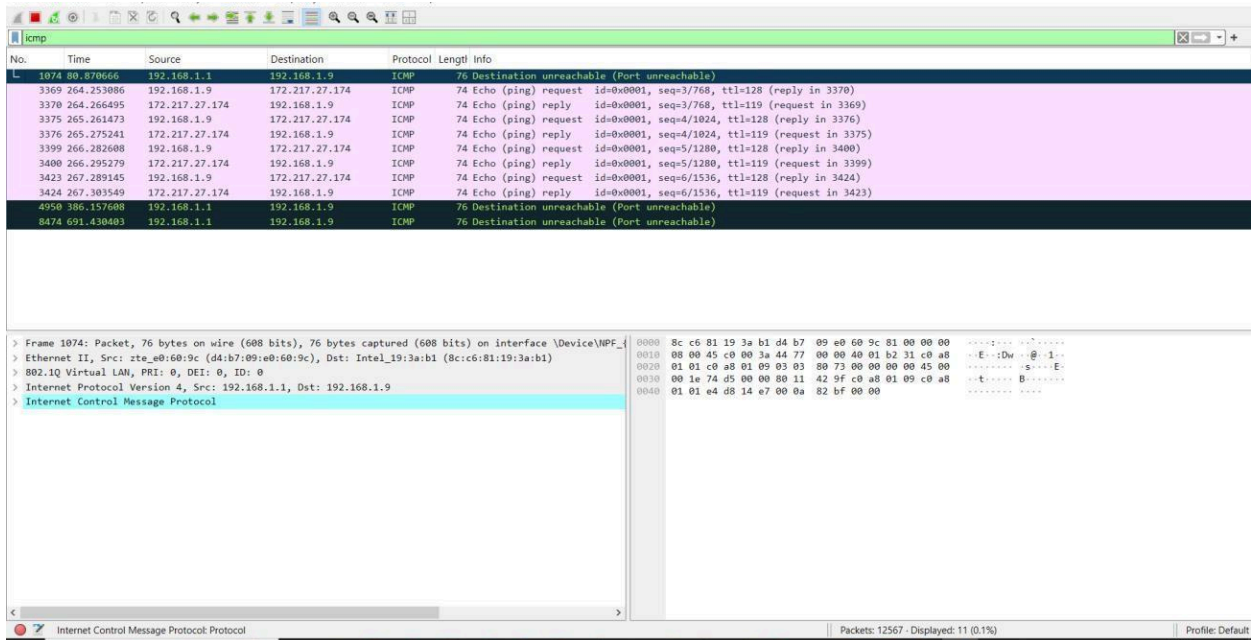
Frame 162: Packet, 591 bytes on wire (4728 bits), 591 bytes captured (4728 bits) on interface \Device\NPF...
 Ethernet II, Src: Intel_19:3a:b1 (8c:c6:81:19:3a:b1), Dst: rte_e0:60:9c (d4:b7:09:e0:60:9c)
 Internet Protocol Version 4, Src: 192.168.1.9, Dst: 142.250.193.46
 User Datagram Protocol, Src Port: 57654, Dst Port: 443
 Data (549 bytes)

Screenshot showing UDP Packets Capture

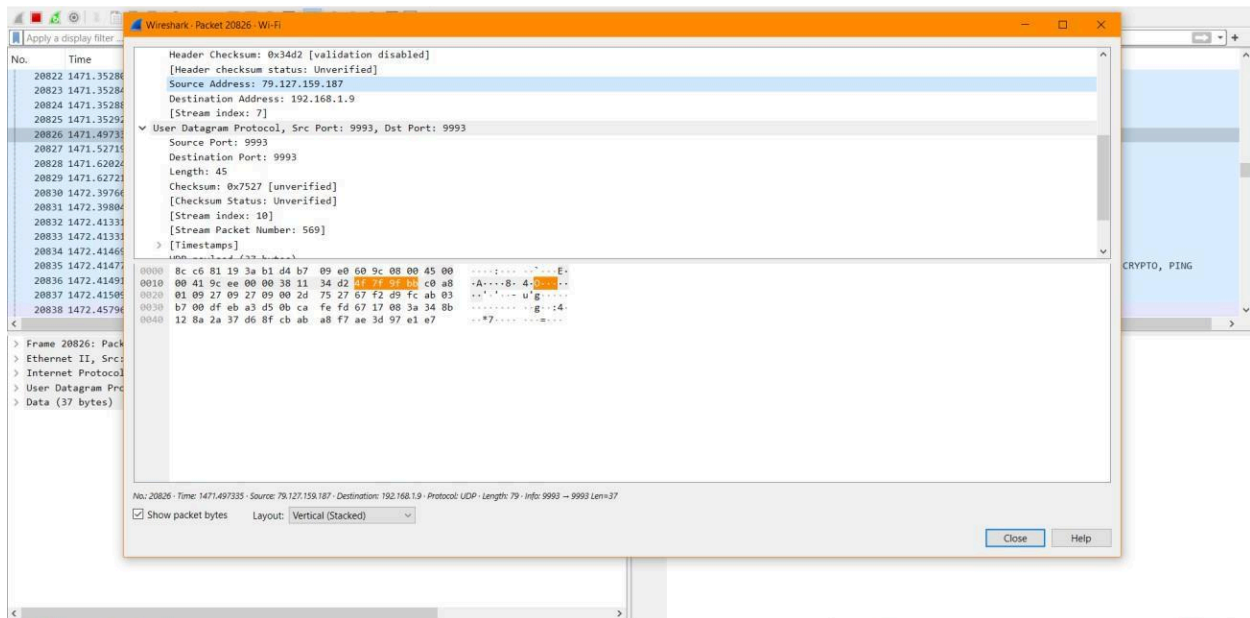
No.	Time	Source	Destination	Protocol	Length	Info
1022	75.980177	192.168.1.9	192.168.1.1	DNS	75	Standard query 0x589b HTTPS ssl.gstatic.com
1023	75.980613	192.168.1.9	192.168.1.1	DNS	75	Standard query 0x9ddf A ssl.gstatic.com
1024	75.994872	192.168.1.1	192.168.1.9	DNS	132	Standard query response 0x589b HTTPS ssl.gstatic.com SOA ns1.google.com
1025	75.996829	192.168.1.1	192.168.1.9	DNS	91	Standard query response 0x9ddf A ssl.gstatic.com A 142.250.193.99
1041	76.086696	192.168.1.1	192.168.1.9	DNS	132	Standard query response 0x589b HTTPS ssl.gstatic.com SOA ns1.google.com
1045	76.098895	192.168.1.1	192.168.1.9	DNS	132	Standard query response 0x589b HTTPS ssl.gstatic.com SOA ns1.google.com
1107	86.717504	192.168.1.9	192.168.1.1	DNS	79	Standard query 0x395f A time.cloudflare.com
1108	86.732204	192.168.1.1	192.168.1.9	DNS	111	Standard query response 0x395f A time.cloudflare.com A 162.159.200.1 A 162.159.200.123
1275	120.078961	192.168.1.9	192.168.1.1	DNS	74	Standard query 0x1bc6 HTTPS www.google.com
1276	120.079280	192.168.1.9	192.168.1.1	DNS	74	Standard query 0x403c A www.google.com
1277	120.092034	192.168.1.1	192.168.1.9	DNS	99	Standard query response 0x1bc6 HTTPS www.google.com HTTPS
1278	120.093536	192.168.1.1	192.168.1.9	DNS	90	Standard query response 0x403c A www.google.com A 142.250.183.196
1347	120.318933	192.168.1.9	192.168.1.1	DNS	70	Standard query 0x7db2 HTTPS google.com
1348	120.319181	192.168.1.9	192.168.1.1	DNS	70	Standard query 0x434f A google.com
1356	120.338013	192.168.1.1	192.168.1.9	DNS	95	Standard query response 0x7db2 HTTPS google.com HTTPS
1369	120.398428	192.168.1.1	192.168.1.9	DNS	86	Standard query response 0x434f A google.com A 172.217.27.174
1449	130.310958	192.168.1.9	192.168.1.1	DNS	87	Standard query 0x29cd A roaming.svc.cloud.microsoft

Frame 85: Packet, 80 bytes on wire (640 bits), 80 bytes captured (640 bits) on interface \Device\NPF...
 Ethernet II, Src: Intel_19:3a:b1 (8c:c6:81:19:3a:b1), Dst: rte_e0:60:9c (d4:b7:09:e0:60:9c)
 Internet Protocol Version 4, Src: 192.168.1.9, Dst: 192.168.1.1
 User Datagram Protocol, Src Port: 59719, Dst Port: 53
 Domain Name System (query)

Screenshot showing DNS Packets Capture



Screenshot showing ICMP Packets Capture



Screenshot showing details of a UDP Packet

Observation: UDP Packet 20826 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 79.127.159.187
- Destination IP Address: 192.168.1.9
- Packet Length: 79 Bytes